

Requirements Traceability Matrix

Smart Lab Equipment & Slot Reservation Portal – covers SRS-SLP-01, SADS-SLP-01, TP-SLP-01.

Chain: **Requirement** → **Use Case** → **Architecture Component** → **Design element** → **Test Case**. ID prefixes: FR / NFR / SEC-OBJ / SEC-REQ (SRS), UC (SRS §5), C-nn (Architecture §4), SD-nn / API-n / DD-nn (Design), TC-nn (Test Plan). API numbers refer to the endpoint catalogue in Design §9.1.

1. Requirement-level matrix

Requirement	Use case	Components	Design elements	Test cases
FR-001	UC-02, UC-03, UC-10	C-01, C-02, C-05, C-11, C-12	SD-01; API-2, API-3, API-8; DD-01, DD-02, DD-03	TC-01, TC-02, TC-03, TC-04, TC-17, TC-21
FR-002	UC-03, UC-09	C-05, C-06, C-11	SD-01 (alt A1); API-8	TC-01, TC-05, TC-13
FR-003	UC-04	C-05, C-11	API-11; DD-05	TC-06, TC-07, TC-23
FR-004	UC-06, UC-12	C-08, C-10, C-11	SD-02; API-12, API-13; DD-07	TC-08, TC-09, TC-22
FR-005	UC-08	C-07, C-10, C-11	API-4, API-5, API-6; DD-06, DD-07	TC-10, TC-21, TC-22
FR-006	UC-01	C-03, C-11	API-1	TC-11, TC-19, TC-20
FR-007	UC-05	C-05, C-11	API-9, API-10; DD-08	TC-12
FR-008	UC-07	C-06, C-10, C-11	API-7; DD-07	TC-13, TC-22
FR-009	UC-11	C-05, C-09	SD-01; DD-04	TC-14
FR-010	UC-03	C-05, C-11	SD-01; DD-02	TC-15
NFR-001	UC-03	C-02, C-05, C-11	SD-01; DD-02, DD-04	TC-16
NFR-002	UC-06, UC-07, UC-08	C-02, C-03, C-04	SD-02; DD-08	TC-18, TC-19
NFR-003	UC-03	C-05, C-11, C-12	SD-01; DD-02	TC-17

2. Security traceability

SEC-OBJ → SEC-REQ → component / control → design mechanism → validation test.

Objective	Requirement	Component(s)	Design mechanism	Test case
SEC-OBJ-01	SEC-REQ-01	C-04, C-02	Role table, deny by default, 403 FORBIDDEN_ROLE; SD-02	TC-18
SEC-OBJ-01	SEC-REQ-02	C-03	Idle-time check, 401 SESSION_EXPIRED; SD-02	TC-19
SEC-OBJ-01	SEC-REQ-04	C-03, C-02	Session required on all but API-1; generic login error; no credentials in logs	TC-11, TC-20
SEC-OBJ-02	SEC-REQ-01, SEC-REQ-05	C-04, C-02, C-11	Validation + parameterised access; DD-02	TC-18, TC-21
SEC-OBJ-03	SEC-REQ-03	C-05, C-04	Ownership check; history filtered by session user; DD-08	TC-12, TC-23
SEC-OBJ-04	SEC-REQ-06	C-10	Audit record in the same transaction; DD-07	TC-22

3. Use case coverage

Use case	Requirements	Components	Diagram / design	Test cases
UC-01 Login / Authenticate	FR-006, SEC-REQ-02, SEC-REQ-04	C-01, C-02, C-03	API-1	TC-11, TC-19, TC-20
UC-02 View Equipment Availability	FR-001	C-01, C-02, C-05	API-2, API-3	TC-01
UC-03 Reserve Equipment Slot	FR-001, FR-002, FR-010, NFR-001, NFR-003	C-02, C-05, C-06, C-09, C-11	SD-01 , API-8	TC-01...TC-05, TC-14...TC-17
UC-04 Cancel Reservation	FR-003, SEC-REQ-03	C-05	API-11	TC-06, TC-07, TC-23
UC-05 View Reservation History	FR-007, SEC-REQ-03	C-05	API-9, API-10	TC-12
UC-06 Return Equipment	FR-004, NFR-002, SEC-REQ-01, SEC-REQ-06	C-02, C-04, C-08, C-10	SD-02 , API-12, API-13	TC-08, TC-09, TC-18, TC-22
UC-07 Update Calibration Status	FR-008, SEC-REQ-01, SEC-REQ-06	C-06, C-10	API-7	TC-13, TC-18, TC-22
UC-08 Manage Equipment Inventory	FR-005, SEC-REQ-01, SEC-REQ-05, SEC-REQ-06	C-07, C-10	API-4...API-6	TC-10, TC-18, TC-21, TC-22
UC-09 Verify Calibration Status	FR-002	C-06	SD-01	TC-05, TC-13
UC-10 Generate Reservation Token	FR-001	C-05	SD-01; DD-01	TC-01
UC-11 Send Confirmation Notification	FR-009	C-09	SD-01; DD-04	TC-14
UC-12 Apply Late-Return Penalty	FR-004	C-08	SD-02 (opt fragment)	TC-08, TC-09

4. Component coverage (no orphan components)

Component	Requirements it serves
C-01 Web Client	FR-001, FR-003, FR-007, FR-009 (display), EIR-01
C-02 API Layer	SEC-REQ-04, SEC-REQ-05, NFR-001, NFR-002
C-03 Authentication & Session Service	FR-006, NFR-002, SEC-REQ-02, SEC-REQ-04
C-04 Access Control Guard	NFR-002, SEC-REQ-01, SEC-REQ-03
C-05 Reservation Service	FR-001, FR-003, FR-007, FR-010, NFR-001, NFR-003, SEC-REQ-03
C-06 Calibration Service	FR-002, FR-008
C-07 Inventory Service	FR-005
C-08 Return & Penalty Service	FR-004
C-09 Notification Adapter	FR-009
C-10 Audit Logger	SEC-REQ-06
C-11 Data Access Layer	NFR-001, NFR-003, SEC-REQ-05
C-12 Relational Database	NFR-003

5. Test case → requirement (reverse)

Test	Requirement(s)	Type
TC-01	FR-001, FR-002	Functional
TC-02	FR-001	Functional / boundary
TC-03	FR-001	Functional / boundary
TC-04	FR-001	Functional / negative
TC-05	FR-002	Functional / negative
TC-06	FR-003	Functional
TC-07	FR-003	Functional / boundary
TC-08	FR-004	Functional
TC-09	FR-004	Functional / boundary
TC-10	FR-005	Functional
TC-11	FR-006, SEC-REQ-04	Functional / security
TC-12	FR-007, SEC-REQ-03	Functional / security
TC-13	FR-008, FR-002	Functional
TC-14	FR-009	Functional / fault injection
TC-15	FR-010	Functional / negative
TC-16	NFR-001	Performance
TC-17	NFR-003, FR-001	Concurrency
TC-18	NFR-002, SEC-REQ-01	Security
TC-19	NFR-002, SEC-REQ-02, FR-006	Security / boundary
TC-20	SEC-REQ-04, FR-006	Security
TC-21	SEC-REQ-05, FR-001, FR-005	Security / negative
TC-22	SEC-REQ-06, FR-004, FR-005, FR-008	Security
TC-23	SEC-REQ-03, FR-003	Security / negative

All 23 tests currently have status *Not Executed*.